

Flow Bridge: Productivity Automation Across Classroom and Career using Agentic AI.

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Abstract --- Introduction- *Flow Bridge is an AI-enabled application that uses a multi-agent based approach to help students and working adults be more productive by helping them organise their work in a structured way, receive intelligent support and gain access to pertinent information quickly. The application consists of multiple agents that work together to help users schedule study times, gather resource materials, address questions, provide student mentors, break down tasks into smaller steps, write emails, schedule meetings and evaluate the productivity of their time spent studying and working. All of these agents have an associated RAG based document support capability that allows users to generate answers based on the documents and syllabus materials the user has uploaded to the application. Therefore, the user's answers will be relevant and clear, as they are all rooted in the same source documents and syllabus materials. To further assist the user in making the transition from academic to professional use, the application employs a modularised design methodology and role-defined agent concept. Flow Bridge has been designed using only local, free (in the sense that no internet connection is required) AI models, making it equally usable in both an academic context (studying) and a professional environment (working).*

Key words: *Multi-Agent System; AI; Retrieval-Augmented Generation; Educational Technology; Productivity Tool; Local LLM; Streamlit.*

I. INTRODUCTION

A. AI-Driven Assistance in Academic and Professional Workflows

The requirements for planning ahead, establishing continuous learning systems, and managing

tasks effectively have been increasing in both academic and professional settings. On the academic side of things, students are expected to keep track of their syllabus, their assessments, and the many resources available to them, while on the professional side of things, the way professionals communicate, schedule meetings, and complete work is done under a strict timeline and within specific budget constraints. Traditional digital tools are not ideal for this environment because they typically do not support all the needs of a student or professional in a single platform. Therefore, the use of AI technology has created the opportunity for creating intelligent agents that have the ability to reason toward a desired outcome and offer assistance in real-time through a personalised approach to each user. Agents within agentic AI are made up of multiple, specialised agents that can work together to solve problems and provide support as each task is being completed. These intelligent agents are currently being utilised in education and productivity areas to move from something that provides only static assistance toward providing users with ongoing, useful interactions that involve pre-empting and explaining issues. Unfortunately, the vast majority of intelligent agent solutions today are primarily cloud-based.

B. Need for Research in Multi-Agent and RAG-Based Productivity Systems

Current educational AI and productivity tools research has typically been focused on planning, tutoring, and communication as distinct functions rather than looking at them collectively to create an integrated multi-agent framework. This isolation of these different types of functions can limit the holistic support available to individuals who work in both the academic and professional worlds. The use of a unified multi-agent framework would help to bring together these types of

functions and would also help to establish more cohesively connected areas of support for end users. RAG can enhance the dependability of a language model's response to a user by anchoring the output to a user-supplied document. It also helps to mitigate hallucinations and increase contextual relevance. Even though RAG is being used within educational applications, a comprehensive integration of Retrieval Augmented Generation (RAG) within a modular, locally deployed, multi-agent architecture is still lacking. Additionally, most approaches to agentic AI have focused on large-scale theoretical models or systems that are too costly and/or difficult to access, creating significant barriers to entry for individuals who cannot afford to purchase these systems. To help bridge these gaps, Flow Bridge explores developing a practical multi-agent system that combines light-weight RAG with role-specific agents to facilitate a more structured planning process and improve access for users.

II. RELATED WORK

The convergence of research on intelligent systems in the area of education and productivity has resulted in research that is closely related. These systems operate at the confluence of multi-agent coordination, personalised learning systems and document-grounded retrieval services. The primary focus of research today is to shift from static, fixed-model-based AI systems to more 'agentic' or autonomous based AI systems; these systems are capable of 'reasoning' (i.e. the ability to draw deductive conclusions) and the execution of multiple steps in a decision-making process. Previous studies have shown that, although the use of multi-agent frameworks has increased scalability for domain-specific applications such as recommender systems and workflow automation, the majority of current multi-agent systems are in prototype forms or only operate in a limited number of domains (i.e. 'single task environments'). Thus, there is a significant gap between current research efforts to develop multi-agent-based platforms that are capable of performing multiple functions/roles (mentoring, professional scheduling) within the same system.

In Fig 1, the addition to the changes in architecture brought about by artificial intelligence (AI), there has been an increased emphasis on adaptive instruction and explanation in academic planning as it relates to the use of educational technological tools based on the LLM Frameworks. The current LLM-based technology has indicated that it can provide students with a more personalised approach to developing their study schedules. Unfortunately, the majority of companies using these types of solutions operate independently from one another, although many students require some sort of integration to provide them with tools that they can use for enhancing productivity during their transition to the world of work. In addition, the use of centralised Cloud APIs creates a level of uncertainty regarding both the privacy of data collected and the cost to institutions, both of which are unclear or at least not widely discussed in current literature. RAG is quickly becoming the leading approach for ensuring that AI-based programs provide valid, factual

information to students, as RAG-type assistants have proven to be successful in reducing the number of errors made (i.e., hallucinations) by these types of systems in Curriculum-Specific contexts. Yet the vast majority of implementations remain limited to how they are designed to work within a single-agent conversational interface.

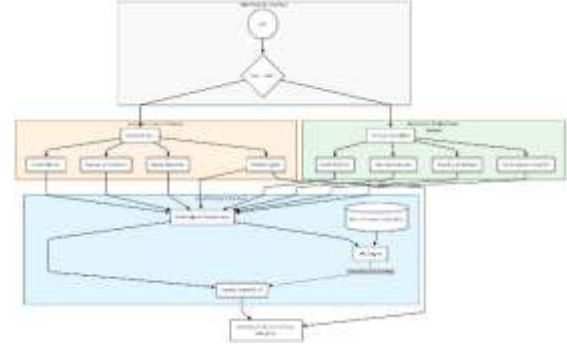


Fig.1. Architecture diagram of Flow Bridge

III. METHODOLOGY

In Fig 2, Flow Bridge architecture is focused on a modular multi-agent architecture based on a localised Retrieval-Augmented Generation (RAG) pipeline. We are focused on specialisation of roles and local implementation in order to mitigate the dual challenges of privacy of data and low cost of computation.

A. System Design and Layered Architecture: The platform is structured into three functional layers: user interaction, agent orchestration and knowledge grounding. This decoupling of these layers will make sure that every functional requirement is handled by a specialised agent, which will lower the cognitive load on the fundamental language model and decrease system complexity. Agents interact through a central state manager instead of having a monolithic structure, exchanging structured context, thus allowing organised decision-making without the dangers of tight coupling. The interface uses a role-based filter, only showing what tools are needed based on either the academic or professional workflow, which enables the system to easily be extended.

B. Agent Roles and Coordination Logic: There are eight specialised agents, which are the exam planners through the task decomposers, with a different objective function that is controlled. Flow Bridge uses shared metadata and intermediate outputs to manage tasks instead of using a high-dependency chain, and causing failure of one agent to bring the system down. This design concept minimises error propagation and limits the operation of each agent to a constricted, quick-answer reasoning space, specific to its own purpose. The strategy is in line with the fact that specialised multi-agent design is more reliable in complex workflow processes compared to general-purpose models.

C. Localised RAG Implementation : To ensure factual integrity, we combined a RAG pipeline, which bases agent response on user-posted documents, e.g. syllabi or technical notes. The pipeline is capable of text extraction, recursive chunking and embedding generation. When a

query is made, the system loads into the LLM context window only the most relevant segments of the documents. Using open-source embedding models and local storing of vectors, we can do so without relying on external API calls, thus dramatically lowering the latency and making sure no sensitive user information exits the local environment.

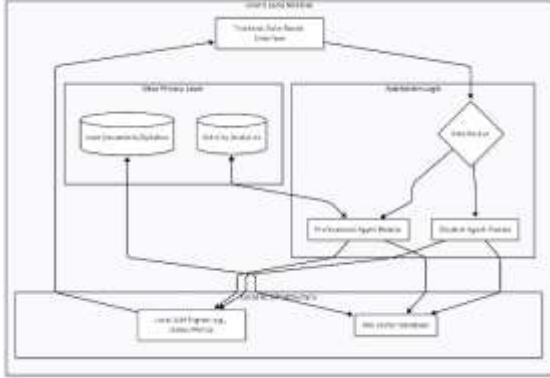


Fig.2. Block Diagram of Flow Bridge

D. Technology Stack and Implementation Workflow The system is developed in Python, and the analysis of the implementation is stipulated by the simplicity and interactivity of the frontend that is developed in streamlit. The entire platform executes the quantised open-source LLM to enable the platform to be used in cost-sensitive or offline systems.

Execution imprisonment is a process which begins with the intent classification: the system receives the query of a user and translates it into the appropriate agent. When the task requires external knowledge (e.g. a "doubt resolution" query), the RAG module is triggered to provide context and the agent builds a final response. This systematic flow ensures that every output is not just explainable, in that it also has a foundation on the parameters of the specific situation of the user either in academics or occupation.

Similarity score (cosine similarity)

$$Score(q, c_i) = \cos(\theta) = \frac{\vec{q} \cdot \vec{c_i}}{\|\vec{q}\| \|\vec{c_i}\|}$$

Select top-k chunks

$$Context = TopK(Score(q, c_i))$$

IV. RESULTS AND DISCUSSION

In Fig 3, The flow bridge platform is considered in terms of its ability to support specialised agent behaviour and document grounding under a resource constrained local environment. As per our findings, the modular architecture comes in handy in addressing the shortcoming of the local LLMs having fewer parameters.

A. Multi- Agent Performance and Task Specialisation. In the case of role specific split up of the system, we discovered that there was a significant increase in the degree of task reliability compared to the base monolithic prompts. Indicatively, the Exam Planner and the Task

Decomposer agents performed their respective objective tasks more articulately with restriction to the functions they were restricted to do. The specialisation was successful at eliminating the scope of reasoning of the local model, without the so-called context drift into which had to creep in general-purpose assistants. In addition, we coordinated, by using our central state manager, rather than chaining the agents, and this coordination served in reducing to a minimum the so-called cascading error effect. Should an agent such as the Resource Collector fail to satisfy a given URL to which the Mentor Agent was competent to provide a status report as per the available metadata

B. Grounded Reasoning with Localised RAG Localised RAG pipeline proved to be pivotal in minimising hallucinated academic problem of doubt-solving. In our experiment, factual coincidence of responses which were founded based on syllabus-specific PDF data had an improved value compared with base model knowledge. Unexpectedly, we have found that lightweight vector storage (e.g. FAISS or ChromaDB) provided us with almost real-time retrieval latency on the consumer-grade hardware, and the availability of cloud infrastructure is not a requirement to successful RAG in personal productivity tools. The chunking strategy continued to have a significant impact on the quality of response however during reading in the document and in very minute chunks a fragmented explanation could be achieved; which is a sacrifice we have made by introducing recursive character splitting.

C. System Deployment and Resource Trade-offs 100 percent data privacy and zero API latency at all times are a brilliant attribute of an institutional deployment of locally hosted models. Although the local models (e.g., Llama-3-8B) will report to time a slight drop in lingo polish compared to proprietary cloud APIs, compounded with role based prompts and document grounding, this weakness of task based workflows was overcome. Streamlit interface brought a light user interface. We also noticed that only a small number of concurrent agents can trigger transient spikes of high value in the resource used by RAM, and, therefore, it may be possible in the future that an agent-queuing system is introduced to facilitate smooth local computation.

D. Limitations and Future Directions. The available documentation on the system is largely dependent on the quality provided by the users that will define the performance of the system at the current times. The correctness of retrieval in the RAG module will also be reduced, when a syllabus or a set of notes is not coordinated and this is typically a bottleneck in a system with grounded AI [13]. We have also been operating our intent routing which involves the usage of a lightweight classifier which is occasionally inaccurate in the classification of the indefinite queries in student and professional domains. Further studies will include more powerful, more semantically motivated routing and

integration of smaller, more locally-related or specialised, (so-called) SLMs to further reduce the hardware requirements of Flow Bridge.

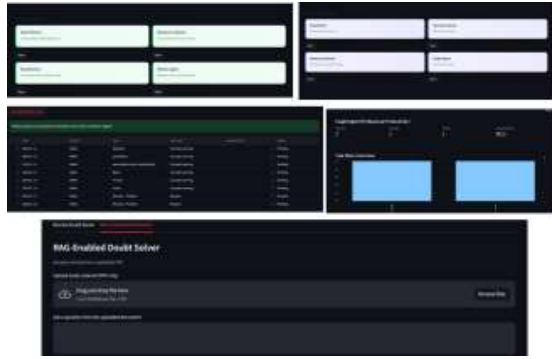


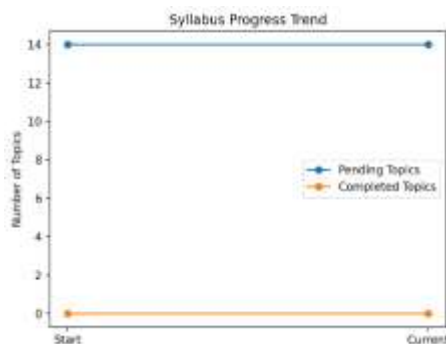
Fig.3. Results of Flow Bridge from different agents

Fig.4. Analytics of Mentor Agent (Student)

V. CONCLUSION AND FUTURE WORK

This paper has detailed the design and implementation of Flow Bridge, an architecture that demonstrates the viability of modular, multi-agent systems in bridging the gap between academic learning and professional productivity. By moving away from monolithic, cloud-dependent AI structures, we have shown that a specialised local agent framework, combined with a lightweight RAG pipeline, can achieve high contextual relevance and factual grounding. Our results suggest that role-based specialisation effectively mitigates the limitations of smaller, locally hosted models, providing a private and cost-efficient alternative for students and professionals alike. Ultimately, Flow Bridge serves as a blueprint for how agentic AI can be deployed in environments where data sovereignty and long-term accessibility are paramount.

Future research will transition from qualitative analysis toward rigorous quantitative benchmarking, specifically measuring task success rates and intent-routing precision across larger user cohorts. We aim to evolve the current rule-based coordination into a dynamic, learning-based orchestration layer that can invoke agents



based on real-time computational availability and task complexity. Furthermore, we plan to expand the RAG mechanism into a multimodal pipeline, enabling the system to parse complex academic diagrams and tables,

which are currently beyond the scope of text-based embeddings. By integrating longitudinal analytics and multi-user collaboration features, we hope to transform Flow Bridge into a comprehensive, adaptive ecosystem for lifelong learning and professional growth.

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